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Q1) What is scanner class in Java? Write code to calculate the area of circle using Scanner class.

The Scanner class in Java is built-in-class in the java.util package that is used to take input from the user.

It can read different types of data such as :

int

float

double

String

long

boolean

It reads input from various sources like the keyboard, (System.in) files or strings.

To use Scanner class, we must import it using:

```
import java.util.Scanner;
```

The Scanner class provides methods like:

nextInt()

nextDouble()

nextLine()

nextFloat()

next()

to read values from the user.

Program to calculate Area of circle using Scanner.
close

```
import java.util.Scanner;
class AreaOfCircle {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter radius of circle");
        double radius = sc.nextDouble();
        double area = 3.14 * radius * radius;
        System.out.print("Area of circle" + area);
        sc.close();
    }
}
```

output:

Enter the radius of circle : 5
Area of circle = 78.5.

Q2) What is an array? Explain different types of array in Java with syntax. Also explain its methods.

Array:

An Array in Java is a data structure used to store multiple values of the same data type in a single variable.

It stores elements in continuous memory location and each element is accessed using index (starting from 0).

Arrays are useful when we need to store and process a fixed number of elements.

Types of Array:

i) One - Dimensional Array :

Used to store list of elements (like marks, price, numbers).

Syntax :

`datatype[] arrayname = new datatype[size];`

ex:

`int[] marks = new int[5];`

ii) Two - Dimensional Array :

Also called matrix or table. It stores data in row and columns.

Syntax :

`datatype[][] arrayname = new datatype[row][col];`

ex:

`int[][] matrix = new int[3][3];`

iii) Multi - Dimensional Array :

Array of arrays (more than 2 dimensions).

Syntax :

`datatype[][][] arrayname = new datatype[x][y][z];`

ex:

`int[][][] data = new int[2][3][4];`

Important Array methods :

Java provides many useful built-in methods in Arrays class. Belong to java.util.Arrays class.

i) `Array.sort()` → Sorts elements in ascending order.

ii) `Array.toString()` → Convert array to readable string format.

iii) `Array.binarySearch()` → Search the values using binary search. (array must be sorted).

iv) `Array.fill()` → Fill the all elements with the same value.

v) `Array.copyOf()` → Copies one array into another.

Q2) Write java code copy contents from one file to another file.

```
import java.io.*;
class FileCopy {
    public static void main(String[] args) {
        try {
            FileInputStream fin = new FileInputStream("1.txt");
            FileOutputStream fout = new FileOutputStream("0.txt");
            int ch;
            while ((ch = fin.read()) != -1) {
                fout.write(ch);
            }
            fin.close();
            fout.close();
            System.out.print("File copied successfully");
        }
        catch (Exception e) {
            System.out.print("Error" + e.getMessage());
        }
    }
}
```

Output:

File copied successfully.

Q4) Explain observable and observer objects with an Example:

In java, the observable-observer model is used to implement the observer design pattern. This mechanism allows one object to notify many other objects whenever a change occurs.

observable object:

- The observable class represents an object that is being watched.
- When its state changes, it notifies all observers.
- observable maintains a list of observers and sends updates to them.

observer object:

- The observer interface represents object that want to be notified of changes in the observable.
- Whenever the observable changes, the Observer update() method is called.

ex:

```
import java.util. Observable;
import java.util. Observer;
class NewAgency extends Observable {
    private String news;
    public void setNews(String news) {
        this.news = news;
        setChanged();
        notifyObservers(news);
    }
}
```

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```
class NewsChannel implements Observer {
    private String news;
    public void update(Observable o, Object args) {
        this.news = (String) args;
        System.out.print("channel received news" + news);
    }
}
```

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```

public class ObserverDemo {
    public static void main(String[] args) {
        NewsAgency agency = new NewsAgency();
        NewsChannel channel = new NewsChannel();
        agency.addObserver(channel);
        agency.setNews("New policy announced");
    }
}

```

output:

channel received news: new policy announced.

Q5). write a java code to create a reading and writing text file.

```

import java.io.*;
class ReadWriteFile {
    public static void main(String[] args) {
        try {
            FileWriter writer = new FileWriter("o.txt");
            writer.write("Hello, this is text file");
            writer.close();
            System.out.println("writing Completed");

            FileReader reader = new FileReader("o.txt");
            BufferedReader br = new BufferedReader(reader);
            String line;
            System.out.print("Reading file Content");
            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }
            br.close();
            reader.close();
        }
        catch (Exception e) {

```



```
System.out.println("Error" + e.getMessage());
```

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output:

writing Completed.

Reading File Content:

Hello, this is a text file.

Q6). Explain Collection Types: ArrayList, LinkedList, Sets, Sequence Map with example.

The collections framework in Java provides different data structure to store and manage groups of object,

It includes interfaces and classes like List, Set, Map, ArrayList, etc.

i) ArrayList.

- ArrayList is resizable array and allow duplicate elements.
- It maintains index-based access making retrieval fast.
- It belongs to List interface.

ex:

```
import java.util.*;
ArrayList<String> list = new ArrayList<>();
list.add("Apple");
list.add("Banana");
list.add("Apple");
```

ii) Linked List.

- Linked List store elements in the form of nodes.
- It's good for insertion and deletion operations.
- It also belongs to the List Interface.

ex:

```
LinkedList<String> names = new LinkedList<>();
names.add("John");
names.add("Sarah");
names.add("Mike");
```

iii) Sets.

- set is collection that does not allow duplicate elements.
- It does not maintain order (in HashSet)
- Common Set: HashSet, LinkedHashSet, TreeSet.

ex:

```
HashSet<Integer> set = new HashSet<>();
set.add(10);
set.add(20);
set.add(10); //duplicate ignored.
```

iv) Sequence.

- Sequence mean elements arranged in linear order and accessed by their position (index) in java [Vector] and [Stack] represents sequence collection.

ex:

```
Vector<String> v = new Vector<>();
v.add("A");
v.add("B");
v.add("C");
```


Map

- Map stores key-value pair.
- key must be unique, but values may be duplicate
- examples: HashMap, TreeMap, LinkedHashMap

ex:

```
HashMap<Integer, String> map = new HashMap<>();  
map.put(1, "Mango");  
map.put(2, "Apple");  
map.put(3, "Banana");
```